Eight audited old/new pairs

Only the initial coefficient basis changed: archived Choi-coordinate draws versus the same normalized draws mapped through a unit-norm Gell-Mann basis. These are eight new runs paired with eight existing endpoints, not sixteen new runs.

Endpoint gate passes: old Choi → new Gell-Mann

Printed adaFGD: 0/4 → 0/4 | FRAMES: 1/2 → 1/2 | Armijo control: 2/2 → 2/2

Method / α	ch	Basis	Fidelity	RMSE	norm. TP	Gate	Status
Q / 5e-06	0	Choi	0.9967170	1.55e-03	7.11e-03	FAIL	complete
		GM	0.9959649	1.71e-03	8.67e-03	FAIL	complete
Q / 5e-06	1	Choi	0.9973493	1.34e-03	5.95e-03	FAIL	complete
		GM	0.9964867	1.51e-03	7.07e-03	FAIL	complete
Q / 5e-04	0	Choi	0.9965359	1.60e-03	3.01e-03	FAIL	complete
		GM	0.9986988	1.05e-03	2.21e-03	FAIL	complete
Q / 5e-04	1	Choi	0.9969312	1.43e-03	2.50e-03	FAIL	complete
		GM	0.9952922	1.75e-03	3.42e-03	FAIL	complete
F / 5e-05	0	Choi	1.0000000	1.11e-06	9.52e-06	FAIL	complete
		GM	1.0000000	1.11e-06	9.52e-06	FAIL	complete
F / 5e-05	1	Choi	1.0000000	8.43e-07	7.95e-06	PASS	complete
		GM	1.0000000	8.43e-07	7.95e-06	PASS	complete
A / 5e-04	0	Choi	1.0000000	1.14e-10	2.07e-10	PASS	converged
		GM	1.0000000	1.33e-10	2.36e-10	PASS	converged
A / 5e-04	1	Choi	1.0000000	1.24e-10	2.22e-10	PASS	converged
		GM	1.0000000	1.44e-10	2.66e-10	PASS	converged

Q = Quiroga's printed adaptive update; F = FRAMES; A = Armijo diagnostic control. α is the fixed TP coefficient per row for Q/A and the initial coefficient for F. Values are rounded for display; gates use unrounded values and terminal status.

Unchanged full gate: fidelity ≥ 0.99, full-design RMSE ≤ 10^{-6}, and normalized TP violation ≤ 10^{-3}; terminal status must be complete or converged. Converged alone is not recovery. Failed terminal runs stay failures.

Only one initializer and one FRAMES sampling stream per channel were tested. This two-channel calibration probe does not select a winner or establish robustness. No validation/held-out channels were used, no gate was relaxed, and no failed result was discarded.

What changed — and what did not

Takeaway. The initializer change did not rescue the printed adaptive rule in these four runs. This does not settle the missing paper conventions or complete the best-for-best comparison. Independent calibration and a frozen validation protocol are still needed before scaling up.

Execution and checks. Ruche array 2038514; all eight archives audited. Full suite: 581 passed, 55 skipped; 9 harness tests passed. Final-factor, native-endpoint and last-checkpoint metrics agree in 8/8 cases. Code merged locally; not pushed.

Frozen setup. Two qubits, rank-one unitary truth, 128 noiseless global-design rows; calibration channel seeds 0 and 1. Every method on a channel shares the same truth, data and mapped initial factor. Old/new pairs share the exact original coefficient draw and all method parameters; only its physical-coordinate basis changes.

Initialization convention. Independent real/imaginary uniform[0,1) coefficients are globally normalized to squared Frobenius norm d=4, then mapped without another normalization. The Hilbert–Schmidt-unit basis orders identity, lexicographic symmetric pairs, lexicographic antisymmetric pairs (-i above, +i below the diagonal), then normalized diagonal generators. The implementation avoids a dense basis-change matrix.

Independent penalties and budgets. Q keeps α fixed (λ=128α), with η scale 1; F starts at $\alpha_0=5\times10^{-5}$ and increases it as $\alpha_0(k+1)^{\wedge}0.25$. Q runs at most 100,000 full updates; F runs 400,000 batch-32 updates, with γ=4/(k+4) and τ=2. Each complete primary run charges 12.8 million nominal optimizer rows, excluding spectral work and checkpoints. Armijo keeps $\alpha=5\times10^{-4}$, has at most 10,000 iterations and charges line-search/terminal-gradient work separately; it is not a matched-work primary competitor.

Interpretation limit. Quiroga's printed adaptive step was not replaced by Armijo. The paper does not fully specify the uniform support, initializer normalization or basis ordering. This explicit Gell-Mann diagnostic therefore is not an exact reproduction. The primary protocol, previous archives and gates remain unchanged; channels 100–107 and 1000–1019 remain outside this probe.

Paper, Section V: https://arxiv.org/html/2312.01311v1#S5

Audit evidence. Both inventories are complete; all eight archive pairs passed source/configuration checks, independent endpoint and gate recomputation, row-accounting checks, and an independent dense Gell-Mann mapping check against the original draw. Every baseline was read from its preserved archive, not rerun.

Provenance / SHA-256

Source commit
`966ef74f57a3e4d9fcd1378df99df3e4641a7d31`

Frozen plan
`cc34b86834f9a6a26a798f8a3d2b39e2de8b850b6564017f87dfbc56d2f016bb`

Complete audit
`73ae52860439868eb12fc59be9dc9f4c15d810e9463e0dc17b0811645a32fdd3`

Individual source/runner/archive hashes and all unrounded endpoints are retained in the audit; supplementary_endpoint_check.json records the additional trace-integrity checks. Audit created: 2026-10-07T18:10:08.210639+00:00